Application Integration/Messaging Services
Amazon SQS
Tutorial

Topics

1. Using Amazon Simple Queue Service (Amazon SQS) via the AWS SDK for Python

Prerequisite

- Complete this tutorial using the AWS Academy Learner Lab dedicated to our module[1].
- Note that if you are completing this tutorial using the AWS Academy Learner Lab, there is a restriction on the **AWS Regions** that can be used, namely all service access is limited to the **us-east-1** and **us-west-2** AWS Regions unless mentioned otherwise.
- This tutorial assumes that you are using the AWS Cloud9 Environment set up for developing in Python. Otherwise please complete the section *Setting up Cloud9 for Cloud Application Development in Python* of the Tutorial on AWS Cloud9 for Python.

Using Amazon Simple Queue Service via the AWS SDK for Python

Amazon Simple Queue Service (Amazon SQS)[2] is a fully managed message queuing service that provides secure, durable, and available queue that enables to integrate and decouple distributed software systems and components.

[1] This assumes that you have accepted the invitation to join the AWS Academy Learner Lab received via your NCI email account.

[2] https://aws.amazon.com/sqs/

Objectives

The aim of this tutorial is to learn how to programmatically use Amazon SQS. More specifically, at the end of this tutorial you should know to perform the following tasks:

- Create a queue
- Send messages to a queue
- Consume messages from a queue
  - Retrieve messages from a queue
  - Delete messages once they have been processed
- Delete a queue

Recall that the AWS SDKs provide two levels of APIs for the AWS services, namely Client (or "low-level") APIs and Resource APIs. Please consult the documentation of Amazon SQS available at https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/sqs.html#sqs.

In this tutorial we will use the SQS Client which is a low-level client representing the Amazon Simple Queue Service (Amazon SQS). Please consult the documentation of Amazon SQS Client available at https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/sqs.html#client.

First, consult the documentation of Amazon SQS Client to identify the available methods which enable us to perform the tasks listed under the *Objectives* section.

Sample Python Program The folder *messaging_sqs_demos* available on the module's Moodle page shows examples of how to perform core operations with Amazon SQS programmatically via the available AWS SDK for Python.

Running the sample Python program In order to run the demo, first, check the available arguments by running the following command at the terminal

```
python messaging_demo.py --help
```

For example, if you would like to both create the queue with the given name e.g. mydemoq, and to delete it once all the operations are performed, you would use both optional arguments, namely

`--create_queue`

and

`--delete_queue`

For example, you would run the following command at the terminal

```
python messaging_demo.py mydemoq --create_queue --delete_queue
```

Resources

- Amazon SQS https://aws.amazon.com/sqs/
- Amazon Simple Queue Service Developer Guide https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/welcome.html
- Amazon Simple Queue Service API Reference https://docs.aws.amazon.com/AWSSimpleQueueService/latest/APIReference/Welcome.html
- Boto3 Amazon SQS Documentation https://boto3.amazonaws.com/v1/documentation/api/latest/reference/services/sqs.html#sqs
- AWS Cloud9 https://docs.aws.amazon.com/cloud9/latest/user-guide/welcome.html
- Python Tutorial for AWS Cloud9 https://docs.aws.amazon.com/cloud9/latest/user-guide/sample-python.html